

# Arithmetic and Order

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- **Definition 1**

The set of **integers** is

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

In this course, we will consider that 0 is not a natural number, that is, the set of **natural numbers** is

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

We endowed the set  $\mathbb{Z}$  with two binary operations

$$\begin{aligned} + : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} & \cdot : \mathbb{Z} \times \mathbb{Z} &\longrightarrow \mathbb{Z} \\ (m, n) &\mapsto m + n & (m, n) &\mapsto m \cdot n \end{aligned}$$

The arithmetic of  $\mathbb{Z}$  is generated by the properties that these operations satisfy.

Notation:  $m \cdot n = mn$ .

- **Properties 1.1**

1. If  $m, n, r \in \mathbb{Z}$  then

$$(m + n) + r = m + (n + r). \quad \text{Associativity of sum}$$

2. There exists a unique element  $0 \in \mathbb{Z}$  such that for all  $m \in \mathbb{Z}$

$$m + 0 = m = 0 + m. \quad \text{Existence of identity for the sum}$$

3. For each  $m \in \mathbb{Z}$  there exists a unique element  $-m \in \mathbb{Z}$  such that

$$m + (-m) = 0 = (-m) + m. \quad \text{Existence of inverse for the sum}$$

4. If  $m, n \in \mathbb{Z}$  then

$$m + n = n + m. \quad \text{Commutativity for the sum}$$

5. If  $m, n, r \in \mathbb{Z}$  then

$$(mn)r = m(nr). \quad \text{Associativity of product}$$

6. There exists a unique  $1 \in \mathbb{Z}$  such that for all  $m \in \mathbb{Z}$

$$m \cdot 1 = m = 1 \cdot m. \quad \text{Existence of identity for the product}$$

7. If  $m, n, a \in \mathbb{Z}$  with  $a \neq 0$  and  $am = an$  then

$$m = n. \quad \text{Cancellation law}$$

8. If  $m, n \in \mathbb{Z}$  then

$$mn = nm. \quad \text{Commutativity for the product}$$

9. If  $m, n, r \in \mathbb{Z}$  then

$$m(n + r) = mn + mr. \quad \text{Distributivity of product}$$

- **Proposition 1.2**

$m \cdot 0 = 0$  for all  $m \in \mathbb{Z}$ .

- **Proof 1.2.1**

$$\begin{aligned} m \cdot 0 &= m \cdot (0 + 0) && \text{Property 2} \\ &= m \cdot 0 + m \cdot 0 && \text{Property 9} \end{aligned}$$

hence

$$\begin{aligned} m \cdot 0 + (-m \cdot 0) &= (m \cdot 0 + m \cdot 0) + (-m \cdot 0) \\ 0 &= m \cdot 0 + (m \cdot 0 + (-m \cdot 0)) && \text{Property 1, 3} \\ 0 &= m \cdot 0 + 0 && \text{Property 3} \\ 0 &= m \cdot 0. && \text{Property 2} \end{aligned}$$

◦ **Proposition 1.3**

For all  $m, n \in \mathbb{Z}$ , if  $mn = 0$  then  $m = 0$  or  $n = 0$ .

■ **Proof 1.3.1**

Let  $m, n \in \mathbb{Z}$  such that  $mn = 0$ . If  $m = 0$ , there is nothing to prove.

If  $m \neq 0$ , then

$$mn = 0 = m \cdot 0. \quad \text{Proposition 1.2}$$

and by Property 7 we have that  $n = 0$ .

◦ **Proposition 1.4**

$-m = (-1) \cdot m$  for all  $m \in \mathbb{Z}$ .

■ **Proof 1.4.1**

Since

$$\begin{aligned} m + (-1)m &= m \cdot 1 + m \cdot (-1) && \text{Property 6, 8} \\ &= m(1 + (-1)) && \text{Property 9} \\ &= m \cdot 0 && \text{Property 3} \\ &= 0. && \text{Proposition 1.2} \end{aligned}$$

therefore  $(-1)m$  is an additive inverse of  $m$  and by the uniqueness of the additive inverse we have  $-m = (-1)m$ .

◦ **Proposition 1.5**

$-(-m) = m$  for all  $m \in \mathbb{Z}$ .

■ **Proof 1.5.1**

Since  $m + (-m) = 0$ , the integer  $m$  is the additive inverse of  $-m$  and by the uniqueness of the additive inverse we have that  $m = -(-m)$ .

◦ **Proposition 1.6 (Rules for signs)**

For all  $m, n \in \mathbb{Z}$ ,

$$(-m)n = -(mn), \quad m(-n) = -(mn), \quad (-m)(-n) = mn.$$

■ **Proof 1.6.1**

$$\begin{aligned} 1. \quad mn + (-m)n &= nm + n(-m) && \text{Property 8} \\ &= n(m + (-m)) && \text{Property 9} \\ &= n \cdot 0 && \text{Property 3} \\ &= 0. && \text{Proposition 1.2} \end{aligned}$$

Therefore  $(-m)n$  is an additive inverse of  $mn$  and by the uniqueness of the additive inverse we have  $(-m)n = -(mn)$ .

$$\begin{aligned} 2. \quad mn + m(-n) &= m(n + (-n)) && \text{Property 9} \\ &= m \cdot 0 && \text{Property 3} \\ &= 0. && \text{Proposition 1.2} \end{aligned}$$

Therefore  $m(-n)$  is an additive inverse of  $mn$  and by the uniqueness of the additive inverse we have  $m(-n) = -(mn)$ .

$$\begin{aligned} 3. \quad & (-m)(-n) = -(m(-n)) && \text{Proposition 1.6(1)} \\ & = -(-mn) && \text{Proposition 1.6(2)} \\ & = mn. && \text{Proposition 1.5} \end{aligned}$$

• **Remark 1**

If  $m^2 = n^2$  it does not imply that  $m = n$ , for example,  $2^2 = 4 = (-2)^2$  but  $2 \neq -2$ . In general

$$0 = m^2 - n^2 = (m - n)(m + n).$$

And by the previous proposition we have that  $m - n = 0 \vee m + n = 0$ , that is,  $m = n \vee m = -n$ .

• **Definition 2**

The **order** in  $\mathbb{Z}$  is defined in the following form

$$m \leq n \iff n = m + k \text{ for some } k \in \mathbb{N}_0.$$

Equivalently,

$$m \leq n \iff n - m \in \mathbb{N}_0.$$

The **strict order** in  $\mathbb{Z}$  is defined in the following form

$$m < n \iff m \leq n \wedge m \neq n.$$

The relation  $\leq$  is a **total order** in  $\mathbb{Z}$ , that is, reflexive, antisymmetric, transitive and any two integers can be compared.

◦ **Properties 2.1**

1. For all  $m, n \in \mathbb{Z}$  either  $m \leq n$  or  $n \leq m$ .
2. If  $m, n, r \in \mathbb{Z}$  with  $m \leq n$  then  $m + r \leq n + r$ .
3. If  $m, n, r \in \mathbb{Z}$  with  $m \leq n \wedge r \geq 0$  then  $mr \leq nr$ .

• **Definition 3**

The **absolute value** of an integer  $m$  is

$$|m| = \begin{cases} m & (m \geq 0) \\ -m & (m < 0) \end{cases}.$$

◦ **Proposition 3.1**

Let  $m, n \in \mathbb{Z}$ ,

1.  $m \leq n$  if and only if  $-n \leq -m$ .
2.  $m^2 \geq 0$ .
3.  $m^2 = 0$  if and only if  $m = 0$ .

■ **Proof 3.1.1**

1. ( $\implies$ ) Suppose that  $m \leq n$  then by Property 2

$$\begin{aligned} m - (m + n) &\leq n - (m + n) \\ -n &\leq -m. \end{aligned}$$

- ( $\impliedby$ ) Suppose that  $-n \leq -m$ , then by Property 2

$$\begin{aligned} -n + (n + m) &\leq -m + (n + m) \\ m &\leq n. \end{aligned}$$

2. If  $m \geq 0$  then by Property 3

$$m^2 = m \cdot m \geq m \cdot 0 = 0.$$

If  $m \leq 0$  then by Proposition 3.1(1)  $-m \geq 0$  and by Property 3 we have

$$m^2 = (-m)(-m) \geq (-m) \cdot 0 = 0.$$

3. ( $\Rightarrow$ ) If  $m^2 = 0$  then  $m \cdot m = m^2 = 0$  and thus  $m = 0$ .

( $\Leftarrow$ ) If  $m = 0$  then  $m^2 = m \cdot m = 0 \cdot 0 = 0$ .